

Coding & Solution

Date	14 July 2026
Team ID	SWTID-2026-7049
Project Name	Emotion Detection & Learning Support Engine (AI Learning Assistant)
Maximum Marks	5 Marks

Coding & Solution

Evaluates the quality and completeness of the implemented code and the overall solution against the requirements defined in the Requirement Analysis phase.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/weeb-with-net/emotion-detection-engine-lsa
Programming Language(s)	Python 3.12.10
Framework(s) Used	Streamlit, TensorFlow (BiLSTM), PyTorch + Transformers (BERT), Google Generative AI SDK / OpenAI SDK via OpenRouter
Key Features Implemented	Dual-model (BiLSTM + BERT) emotion detection; data-driven trust/decision engine; mixed-emotion detection; LLM-based response generation with template fallback; opt-in CSV interaction logging; Plotly analytics dashboard (Emotions + Fields tabs); graceful degradation via DISABLE_BERT / DISABLE_AI_RESPONSE.
Pending / Incomplete Features	CSV-based retrieval prediction (UI toggle present, disabled, no backing pipeline); bring-your-own LLM API key (dropped — would need the LLM clients refactored off their module-level singleton pattern); Analytics "Summary" tab (scaffolded, intentionally empty, no scope defined yet).
Setup / Run Instructions	See setup.md: clone the repo, create a Python 3.12 venv, install PyTorch (GPU build if available), then pip install -r requirements.txt, add a .env with GEMINI_API_KEY / OPENROUTER_API_KEY, then run: streamlit run app.py

Code Quality Checklist

S.No	Criteria	Status
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes — 10/10 automated end-to-end tests passing with real weights and real LLM calls
6	Code is committed to a version control repository	Yes — 46 commits on GitHub

Additional Notes / Comments

The public deployment intentionally runs a reduced feature set (BiLSTM-only, template responses) due to an unresolved OpenRouter-call hang specific to the Streamlit Cloud environment. Full-feature behavior (both models, real LLM responses) is verified locally and documented in VALIDATION_REPORT.md — this is a deliberate, disclosed tradeoff for the public link, not a defect.